Scenario

Review the scenario below. Then complete the step-by-step instructions.

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization’s network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- **Identify** security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- **Protect** internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.

- **Detect** potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- **Respond** to contain, neutralize, and analyze security incidents; implement improvements to the security process.

Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Step-By-Step Instructions

Follow the instructions and fill in the sections to complete the activity. Then, go to the next course item to compare your work to a completed exemplar.

Step 1: Access the incident report analysis template

To access template for this course item, click the following link and select *Use Template*.

Link to template:

- Incident report analysis

Link to supporting materials:

- Applying the NIST CSF
- Example of an incident report analysis

OR

Step 2 : Summarize the security event

> Using the template provided, provide a summary of the security event that occurred. Include information about the security event, its cause, the impact, and the response. You can also include information about targeted systems, the attack source, and the estimated impact.

Step 3 : Identify the type of attack and the systems affected

Think about all of the concepts covered in the course so far and reflect on the scenario and define what type of attack occurred and which systems were affected. List this information in the incident report analysis worksheet in the section titled “Identify.”

step 4: Protect the assets in your organization from being compromised

Next, you will assess where the organization can improve to further protect its assets. In this step, you will focus on creating an immediate action plan to respond to the cybersecurity incident. When creating this plan, reflect on the following question:

- What systems or procedures need to be updated or changed to further secure the organization’s assets?

Write your response in the incident report analysis template in the “Protect” section.

Step 5 : Detect similar incidents in the future

It is important to continuously monitor network traffic on network devices to check for suspicious activity, such as incoming external ICMP packets from non-trusted IP addresses attempting to pass through the organization’s network firewall.

step 6 : Respond to future cybersecurity incidents

For this step, consider ways you and your team can monitor and analyze network traffic, software applications, track authorized versus unauthorized users, and detect any unusual activity on user accounts. Write your response in the incident response analysis worksheet in the “Detect” section.

step 7 : Recover from the incident

After identifying the tools and methods you and your organization have in place for detecting potential vulnerabilities and threats, create a response plan in the event of a future incident. This typically happens after the incident occurred and has been resolved by you and your team. In this case, you will create a response plan for future cybersecurity incidents. Some items to consider when creating a response plan to any cybersecurity incident:

- How can you and your team contain cybersecurity incidents and affected devices?
- What procedures are in place to help you and your team neutralize cybersecurity incidents?
- What data or information can be used to analyze this incident?
- How can your organization's recovery process be improved to better handle future cybersecurity incidents?

Write your response in the incident report analysis template under the "respond" section.

Consider what steps need to be taken to help the organization recover from the cybersecurity incident. Reflect on all the information you gathered about the incident in the previous steps to consider which devices, systems, and processes need to be restored and recovered.

Consider the following questions:

- What information do you need to be able to recover immediately?
- What processes are in place to help the organization recover from the incident?

Write your response in the "recover" portion of the worksheet.